EXPLAINABLE MEDICAL AI DIAGNOSTIC REPORT

Automated Chest X-Ray Screening for Pneumonia Detection with Visual Explanations

Patient Name:	Anonymous Patient	**Report Date:**	2026-07-01 01:47:41
Diagnostic ID:	REC-#4	**Model Version:**	PneumoniaCNN v1.0.0

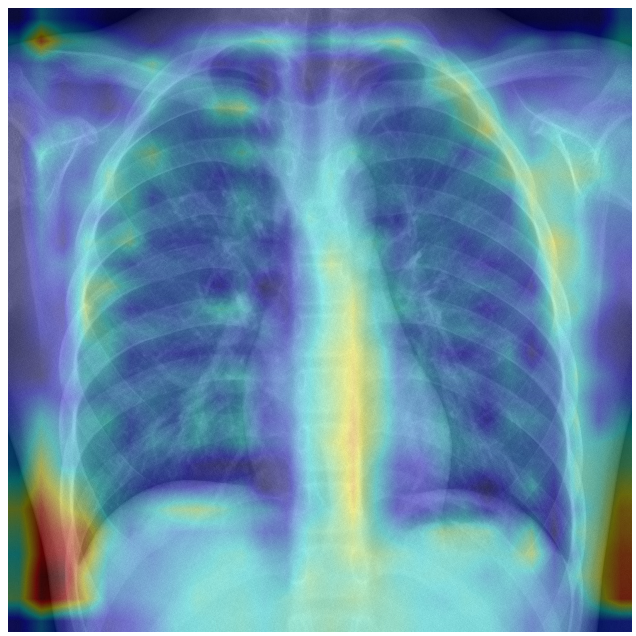
AI DIAGNOSTIC RESULT: PNEUMONIA
Probabilities — Pneumonia: 50.45% | Normal: 49.55%

Confidence score: 50.45%
Analysis Processing Time: 5.32 seconds

IMAGE FINDINGS & EXPLAINABLE AI ANALYSIS

Original Chest X-Ray

Grad-CAM Localization Heatmap



AI DIAGNOSTIC FINDINGS CHECKLIST

✓ **Lung Opacity: The model detected areas of increased density in pulmonary fields.**
✓ **Lower Lobe Abnormalities: Focal densities and consolidations localized in the lobes.**
✓ **High Density Infiltrates: Fluid density regions detected, strongly shifting probability values.**

Visual Explanation Summary: The AI model identified localized regions of high density and opacity, highlighted by the red/yellow regions in the Grad-CAM heatmap above. These patterns correspond to consolidated lung parenchyma and fluid infiltrates, which are standard clinical indicators of pneumonia infection.

Disclaimer: *This diagnostic report was generated by an experimental Artificial Intelligence system. It is intended solely for educational, portfolio demonstration, and decision-support purposes. This report does NOT constitute a medical diagnosis. The findings and explanations should always be verified by a board-certified physician or radiologist.*